10 The table shows some information about eight planets.

Planet	Distance from Earth (km)	Mass (kg)
Earth	0	5.97×10^{24}
Jupiter	6.29×10^{8}	1.898×10^{27}
Mars	7.83×10^{7}	6.42×10^{23}
Mercury	9.17×10^{7}	3.302×10^{23}
Neptune	4.35×10^{9}	1.024×10^{26}
Saturn	1.28×10^{9}	5.68×10^{26}
Uranus	2.72×10^{9}	8.683×10^{25}
Venus	4.14×10^{7}	4.869×10^{24}

(a) Write down the name of the planet with the greatest mass.

The largest mass is 1.898×10^{27} kg.

Jupiter

1 mark
Largest power of ten in the mass column

(1)

(b) Find the difference between the mass of Venus and the mass of Mercury.

Mercury: $3.302 \times 10^{23} = 0.3302 \times 10^{24}$

Difference $= (4.869 - 0.3302) \times 10^{24}$

$= 4.5388 \times 10^{24}$ kg

1 mark
Subtract masses using the same power of ten

4.5388×10^{24} kg

(1)

Nishat says that Neptune is over a hundred times further away from Earth than Venus is.

(c) Is Nishat right?
You must show how you get your answer.

$100 \times$ Venus distance $= 100 \times 4.14 \times 10^{7}$

$= 4.14 \times 10^{9}$ km

1 mark
Compare Neptune with 100 times Venus

Yes. $4.35 \times 10^{9} > 4.14 \times 10^{9}$.

1 mark
Correct comparison

(2)

(Total for Question 10 is 4 marks)

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